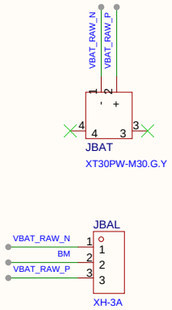
DesktopAssistant Integrated | Battery protection and balance / 电池保护与均衡

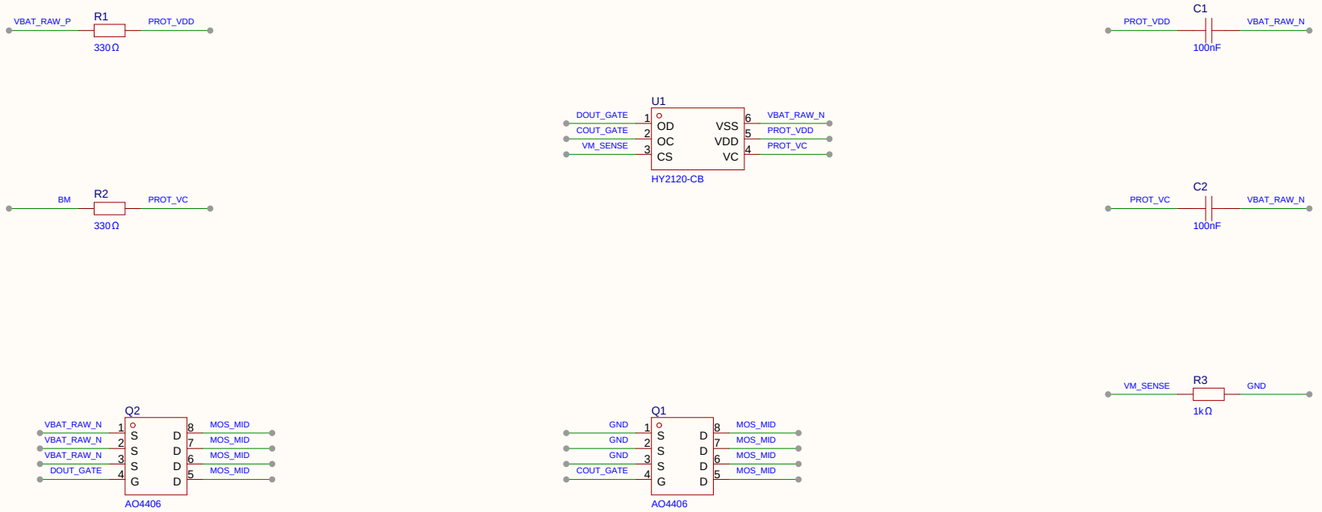
Revision 1.1 2026-09-12 01 / 05

2S battery input / 原始电芯端



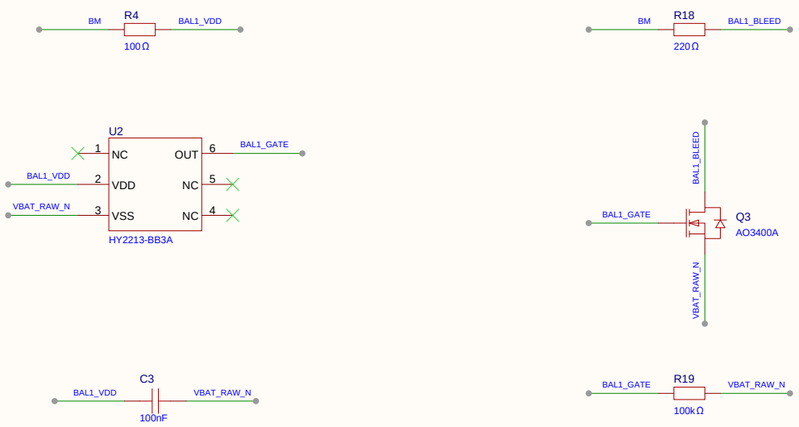
JBAL: 1=B , 2=BM, 3=B+; verify cable polarity.

HY2120-CB protection / 过充过放与短路保护

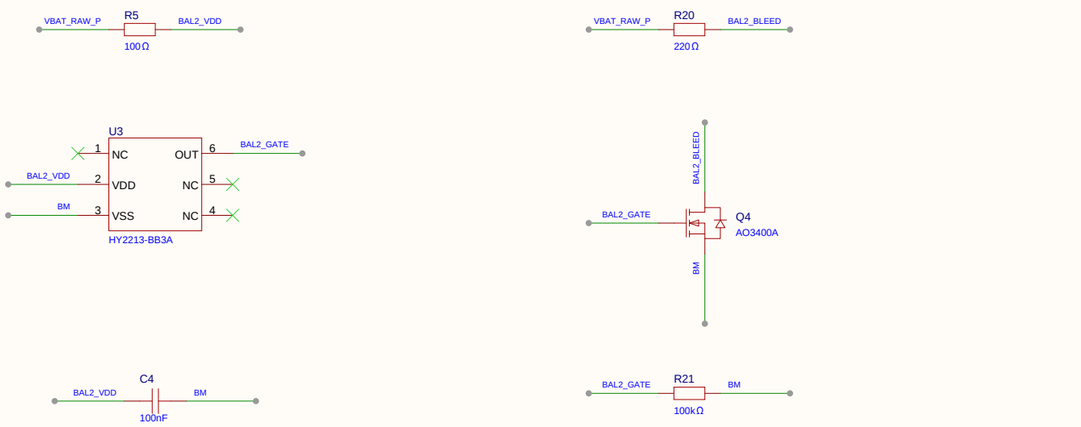


B → Q2(S-D) → MOS_MID → Q1(D-S) → protected GND

Lower cell balance / 下节电芯



Upper cell balance / 上节电芯



Balance current ≈ 19mA at 4.2V; each bleed resistor ≈ 0.08W.

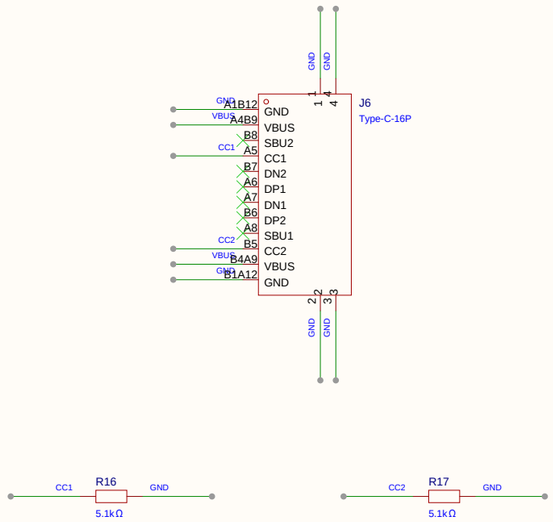
GND is protected pack negative. Do not short GND to VBAT_RAW_N.

Protection and balance thresholds require bench validation before battery use.

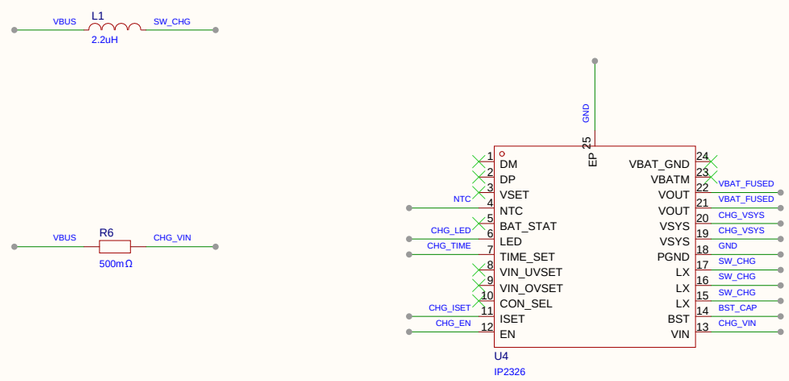
DesktopAssistant Integrated | USB-C charger / 2S 锂电充电

Revision 1.1 2026-09-12 02 / 05

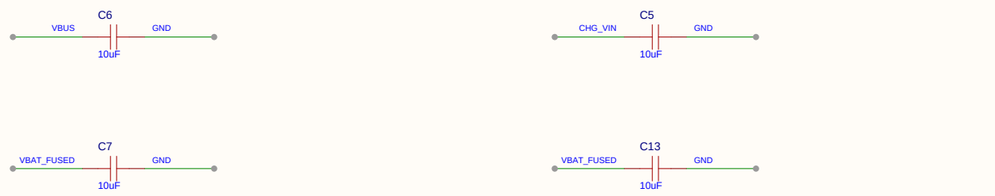
5V USB-C input / 专用 5V 适配器



IP2326 synchronous boost charger



Input / output decoupling



Charge enable and status / 默认禁止充电



Charge settings / 电流、超时和电池温度



2S: CON_SEL NC; 8.4V: VSET NC; VIN UV/OV: NC defaults.

External HY2213 balance: IP2326 pins 23/24 left NC.

Use regulated 5V ≥ 2A adapter. D+/D unused; no QC request.

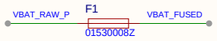
JP_CHG defaults OPEN. NTC must touch battery; charge with main switch OFF.

0.353A is calculated; validate charge current/termination before battery use.

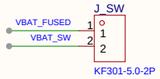
DesktopAssistant Integrated | Power and MCU / 配电与主控

Revision 1.1 2026-09-12 03 / 05

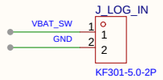
Protected pack → fuse → switch → regulators

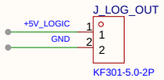


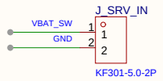
Fit 029707.5WXNV 7.5A fuse into F1.

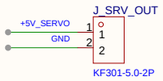


1=fused input; 2=switched output.



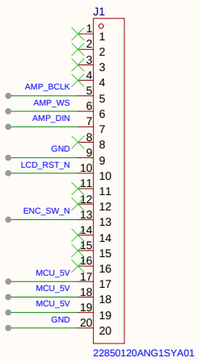


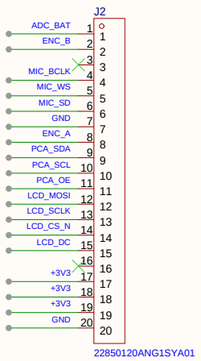




Two external regulator modules; outputs are separate. Set both to 5.0V before connection.

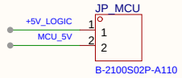
ESP32-S3 carrier sockets / 保留已核对的开发板针序

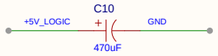




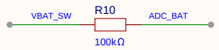
MCU 5V isolated by JP_MCU; OPEN during USB debugging.

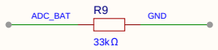
Logic supply / USB isolation

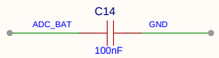




Battery ADC / switched measurement







ADC_BAT ≈ 2.08V at pack 8.4V.

Each module connector: pin 1 positive, pin 2 GND. Modules are external, not on-board footprints.

J_SW is a cable interface, not a diode. Use a DC-rated switch appropriate for servo startup current.

DesktopAssistant Integrated | Display, audio and controls / 显示音频与交互

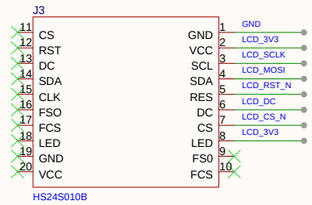
Revision 1.1 2026-09-12 04 / 05

3.3V LCD supply / 独立隔离的屏幕支路



Select ONE LCD source. Keep JP_LCD open when J10 is used.

SPI display / HS24S010B



Pins 1-10 = primary interface; alternate pins 11-20 unused.

I2S module harnesses / 音频线束接口



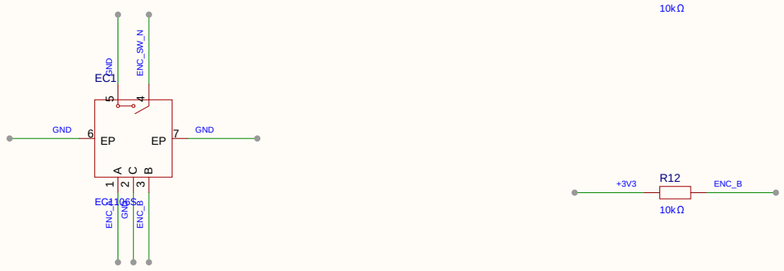
J4: 1=BCLK, 2=WS, 3=SD, 4=3V3, 5=GND, 6=L/R GND.

J5: 1=BCLK, 2=WS, 3=DIN, 4=5V, 5=GND, 6=GND aux.

These are wire-harness pinouts; follow signal names on the modules.

Speaker connects to amplifier OUT+/OUT , not motherboard GND.

EC11 controls / 旋钮与按压



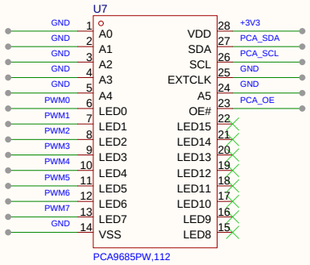
All ESP32 signals are 3.3V. Debounce encoder and button in firmware.

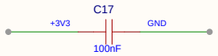
LCD source follows RevB isolation strategy. This page does not connect the two regulator outputs.

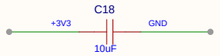
DesktopAssistant Integrated | PWM and servos / 舵机控制与配电

Revision 1.1 2026-09-12 05 / 05

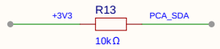
PCA9685 " I2C 0x40 " OE high = outputs disabled

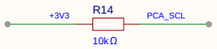


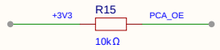




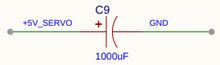
I2C pull-ups and safe startup

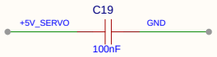




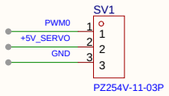


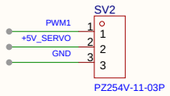
OE defaults HIGH. Firmware must initialize PWM before enabling.

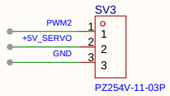


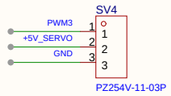


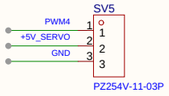
Eight 3-pin servo headers / 1=PWM, 2=+5V_SERVO, 3=GND

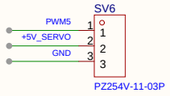


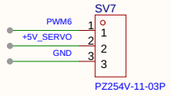


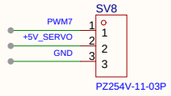












Servo power has its own 5V converter; route high-current return directly to protected power ground.

LED8-LED15 unused. EXTCLK tied to GND for internal oscillator. All address pins are GND.

No servo motion until supply, polarity, current limit and firmware fail-safe behavior have been tested.